

Wheat Price Forecasting using Machine Learning

Using Real Mandi Data (Haryana)

Presented By : Chirag Garg
Course : B.Tech (Data Science)
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Sap Id : 500121978

Introduction

1. Agriculture is a major part of the Indian economy.
2. Crop prices change daily due to demand, supply, weather, and market conditions.
3. Farmers and traders often face difficulty in deciding the right time to sell crops.
4. This project uses Machine Learning to predict future wheat prices based on historical data.
5. This project demonstrates the practical application of data science in solving real-world agricultural problems.

Problem Statement

- Crop prices are unpredictable
- Farmers lack proper market insights
- Wrong timing can lead to financial loss
- There is a need for a system that can analyze past data and predict future prices.
- A data-driven prediction system can help reduce uncertainty in crop pricing.

Objective

- To analyze historical wheat price data
- To build a predictive model using ML
- To compare different models
- To provide future price estimation
- The goal is to assist in better decision-making.
- The system aims to convert raw data into meaningful insights for real-world use.

Dataset

1. The dataset is collected from government sources:

- AGMARKNET
- data.gov.in

2. It includes real mandi data of Haryana.

3. Features:

- Arrival Date (time factor)
- Modal Price (target variable)
- Arrival Quantity (supply factor)

4. Data is real-world and reliable.

5. The dataset reflects actual market transactions, making the model realistic and practical.

Data Preprocessing

1. Steps performed:

- Removed missing and null values
- Converted price values from string to numeric
- Converted date into datetime format
- Created a new feature “Day” for model training
- Cleaned arrival quantity

2. Proper preprocessing improves model performance.

3. Clean and structured data helps the model learn accurate patterns.

Models Used

1. Different ML models were applied:

- Linear Regression
- Multiple Linear Regression
- Random Forest Regressor

2. Random Forest is an ensemble model which combines multiple decision trees for better accuracy.

3. It is capable of handling non-linear relationships and improves prediction performance.

Model Evaluation

1. To evaluate model performance, MAE was used:

Mean Absolute Error (MAE):

Measures average error between actual and predicted values

Lower MAE = Better model

2. Result:

- Linear Regression → Higher error
- Random Forest → Lower error

3. due to its lower error and better accuracy.

Result

1. Model can predict wheat price for any given date
2. It can generate:
 - a. Daily predictions
 - b. Weekly predictions
 - c. Yearly forecast
3. The prediction helps in understanding price trends.
4. It was observed that price trends are influenced by arrival quantity, showing a supply-demand relationship.

Conclusion & Future Scope

- Machine Learning can be used for price prediction
- Random Forest gives better results
- Model works well on real mandi data
- The model successfully converts historical data into actionable insights.

THANK YOU